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Seat No. _____



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SARDAR PATEL UNIVERSITY

Master of Science (Information Technology) (Semester I) Examination-2022

PS01CINT55: Operating System Concepts

Date: 14/11/2022, Monday

Total Marks: 70

Time: 10.00 AM - 01.00 PM

NOTE:

1. Figure to the right indicate full marks of the questions

Q.1 Multiple Choice Questions.

[08]

1. The number of Processes completed per unit of time is called as
(A) Efficiency (B) Capacity
(C) Throughput (D) Output
2. FIFO scheduling is
(A) Pre-emptive Scheduling (B) Fair Share Scheduling
(C) Non Pre-emptive Scheduling (D) Deadline Scheduling
3. Semaphore is a/an _____ to solve the critical section problem.
(A) Hardware (B) Integer variable
(C) Firmware (D) Program
4. What is the minimum number of Processes that can have a deadlock amongst themselves?
(A) 1 (B) 2
(C) 3 (D) 4
5. _____ is a memory management technique in which process address space is broken into blocks of the same size called pages.
(A) Fragmentation (B) Low Memory
(C) Paging (D) None of these
6. A computer can address more memory than the amount physically installed on the system. This extra memory is actually called _____.
(A) Fragmentation (B) Paging
(C) Virtual Memory (D) None of these
7. A _____ is a container that is used to contain folders and files.
(A) Directory (B) Disk
(C) Stack (D) None of these
8. RAID stands for _____.
(A) Read And Input Disk (B) Redundant Array of Independent Disk
(C) Read And IO Disk (D) None of these

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(P.T.O.)

Q.2 Answer the Following. (Attempt Any SEVEN out of NINE) [14]

- a. What is BIOS? What is its role?
- b. Explain the CPU Burst Time, Response Time and Turn Around Time with respect to Process Scheduling.
- c. What are short, long and medium-term scheduling?
- d. Differentiate between multi-tasking and multi-processing systems.
- e. What is busy waiting?
- f. What is fragmentation? List out types of fragmentation.
- g. Define virtual memory.
- h. Define Seek time in terms of disk scheduling.
- i. What is file? List out various operations performed on the file.

Q.3 [12]

- (A) Write a short note on types of Operating Systems.
- (B) Explain interrupts. Give two examples where interrupts are raised and would be handled by an Operating system.

OR

- (B) Describe the importance of Command line interface and Graphical user interface of Operating systems, each giving pros and cons of both through an example.

Q.4 [12]

- (A) Define the concept of 'Process'. List the possible contents of PCB. Draw simple process State –Transition diagram and explain various process states through which a process may move during its lifetime.
- (B) What are semaphores? How semaphores are implemented. Write a solution to producer consumer problem using semaphores.

OR

- (B) Discuss the various conditions that lead to deadlock. Explain deadlock detection and recovery with appropriate example.

Q.5 [12]

- (A) What is page replacement algorithm? Explain optimal page replacement algorithm technique with suitable example.
- (B) Write a note on swapping in context of memory management.

OR

- (B) Write a note on demand paging technique.

Q.6 [12]

- (A) Explain FCFS disk scheduling algorithms with example.
- (B) Explain Directory Structures in Operating System.

OR

- (B) What is RAID? Explain RAID structures in detail.